



DIGITAL TECHNOLOGY

Academic Year 2024-25

Prof. Barbara PERNICI

Exam 10/2/2026

Total time 1.30 h

Last name

First name

Matricola/Person code

Signature

Please remember that:

- The exam is closed books.
- The use of cellular phones or any other electronic devices during the exam is forbidden.

It is necessary to answer at least partially all the questions for a positive evaluation.

Given answers should be explained, lists of bullet items are insufficient to answer a question.

Please write the answer for each question on a separate piece of paper.

Question 1 [11 points]

Explain the main characteristics of the Agile methodologies and discuss how the traditional project Manager responsibilities split into the different roles in the Scrum framework.

Question 2 [11 points]

Given the following time series for insurance claims, identify possible quality problems and describe them, as well as possible corrections:

Id	Name	Location	Issue	Date
1	Brian Muller	Milan	Health	21/1/2026
2	Rose Faber	Milano	Pneumonia	21/1/2026
3	Brian Mulle	Rome	Health	
4	Charles White	Milan	Hospitalization	21/1/2026
5	Brian Muller	Rome	Health	21/1/2026
6	Fabian Brown	Venice	Accident	22/2/2027

[please turn over]

Question 3 [11 points]

Consider the following fragment of Python code:

```
items = ["pen", "book", "notebook", "eraser"]
prices = [2, 10, 5, 1]

data = []

for i in range(len(items)):
    data.append((items[i], prices[i]))

def adjust_price(p):
    if p > 5:
        return p - 1
    else:
        return p + 1

adjusted = []

for item, price in data:
    adjusted.append(adjust_price(price))
```

Questions:

- 1) What is the content of the `adjusted` variable after running all the code?
- 2) Store the content of `adjusted` in a dictionary.
- 3) Using Pandas, calculate the percentage change between the original price and the adjusted price.